SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 5

# Cost–Volume–Profit, Variable Costing & Activity–Based Costing

Managerial accounting answers three questions: what happens when volume changes (CVP)? which costs are product costs (variable vs absorption)? which products/customers actually consume resources (ABC)? Includes Java Roasters, Classic Brass, Meridian and Blue Harbor exercises.

## Material

### 1 CVP: five moving parts and contribution margin

CVP links price, variable cost, fixed cost, volume and sales mix. `CM = Sales - Variable costs` covers fixed costs first, then becomes profit. Racing Bicycle: price $500, VC $300, unit CM $200, **CM ratio 40%**, fixed $80k; each extra $1 of sales adds $0.40 CM.

### 2 Break-even, target profit, margin of safety, operating leverage

Break-even = Fixed ÷ unit CM = 400 bikes = Fixed ÷ CM ratio = $200k — a floor, not a goal. Target profit units = (Target + Fixed) ÷ unit CM = 900 bikes.

**Margin of safety** = actual – break-even sales (e.g. $12,500 = 12.5%).

**DOL** = CM ÷ operating income; %ΔNOI = DOL × %ΔSales. High fixed costs (DOL 4.0): profit $8k–$72k on ±20% sales; low fixed (DOL 2.5): $20k–$60k. Leverage cuts both ways.

**Sales mix:** weighted CM per bundle ($70) → 5,000 bundles = 10,000 A + 15,000 B; a shift to low-CM products raises break-even.

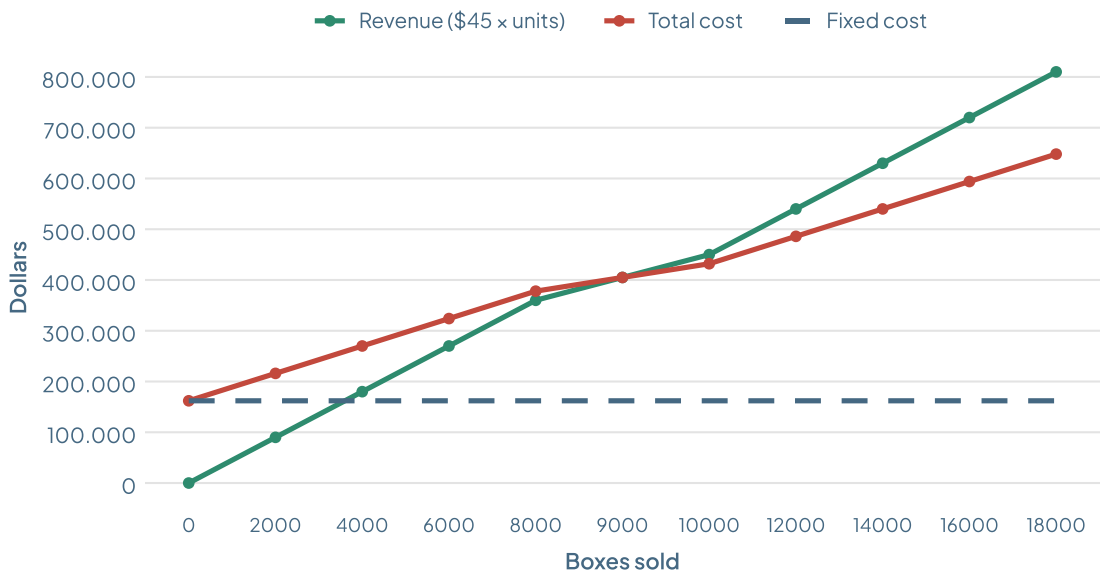
**Assumptions:** constant price, linear costs, constant mix, no inventory change. Use CVP as a model, not a guarantee.

## 3 See it: the CVP graph and a break-even calculator

Deep dive

Revenue starts at zero with slope = price; total cost starts at fixed cost with slope = variable cost. Where they cross is break-even.

**Java Roasters CVP graph**




Lines meet at 9,000 boxes ($405,000); at 12,000 boxes the gap is the $54,000 profit.

Try it with Java Roasters' numbers ($45 price, $27 variable cost, $162,000 fixed cost).

**Reading DOL**

DOL = total contribution ÷ profit; it rises sharply near break-even, so profit becomes very sensitive.

## 4 Variable vs absorption costing, and segment reporting

*Listed in this week's map; summarized from standard managerial accounting.*

The only difference is **fixed manufacturing overhead**: absorption costing inventories it (external reporting); variable costing expenses it (internal decisions, consistent with CVP). When production exceeds sales, absorption profit is higher — a lever that can be abused.

**Segment reporting:** segment margin = segment CM – traceable fixed costs; common costs remain if a segment closes, so judge segments on segment margin.

## 5 Why variable and absorption costing profits differ

Deep dive

The only difference is fixed manufacturing overhead (FMOH): absorption costing stores part of it in inventory when production exceeds sales; variable costing expenses it all.

| Condition | Higher profit |
|---|---|
| Production > sales | Absorption |
| Production < sales | Variable |
| Equal | Same |

```
Profit difference = FMOH per unit × change in inventory units
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**Example**

FMOH $10/unit, inventory up 4,000 units → absorption profit $40,000 higher.

**Example: absorption vs variable profit as production changes**

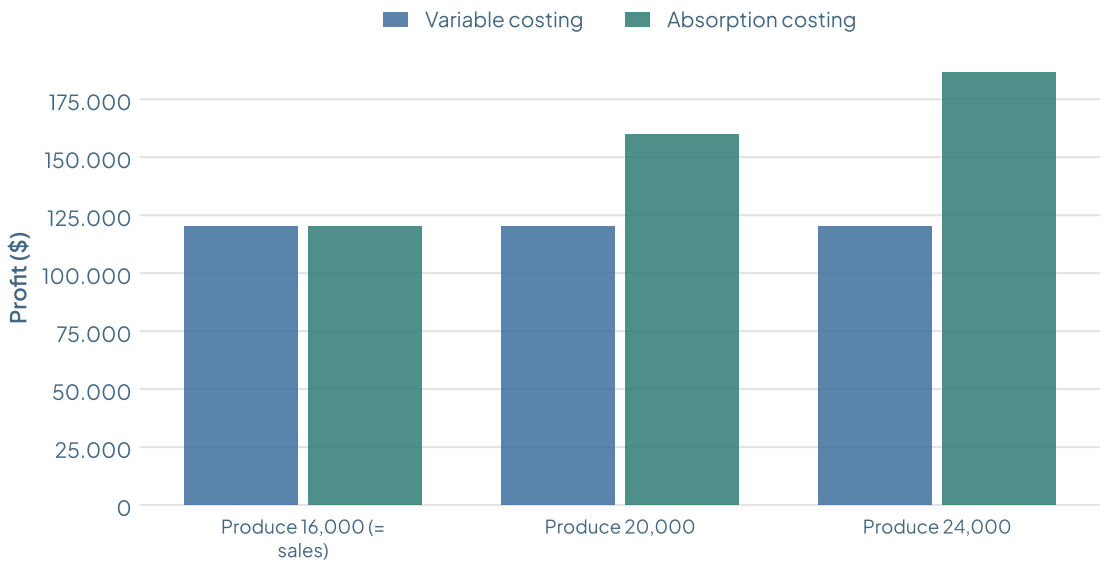



Illustration: sales fixed at 16,000 units, FMOH $200,000. More production parks more FMOH in inventory.

! **Watch out**

Absorption profit can be "boosted" by overproducing.

## 6 Activity-Based Costing (ABC)

Traditional systems use one rate; ABC asks what activities cause cost: resources → activities → cost objects. ABC assigns some nonmanufacturing costs, excludes organization-sustaining costs, and uses many pools. Hierarchy: unit, batch, product, customer, organization-sustaining. Five steps: define activities, assign costs to pools, compute rates, assign to objects, report. Limitations: cost, resistance, misinterpretation, non-GAAP, subjective allocations — always ask whether the cost will actually change.

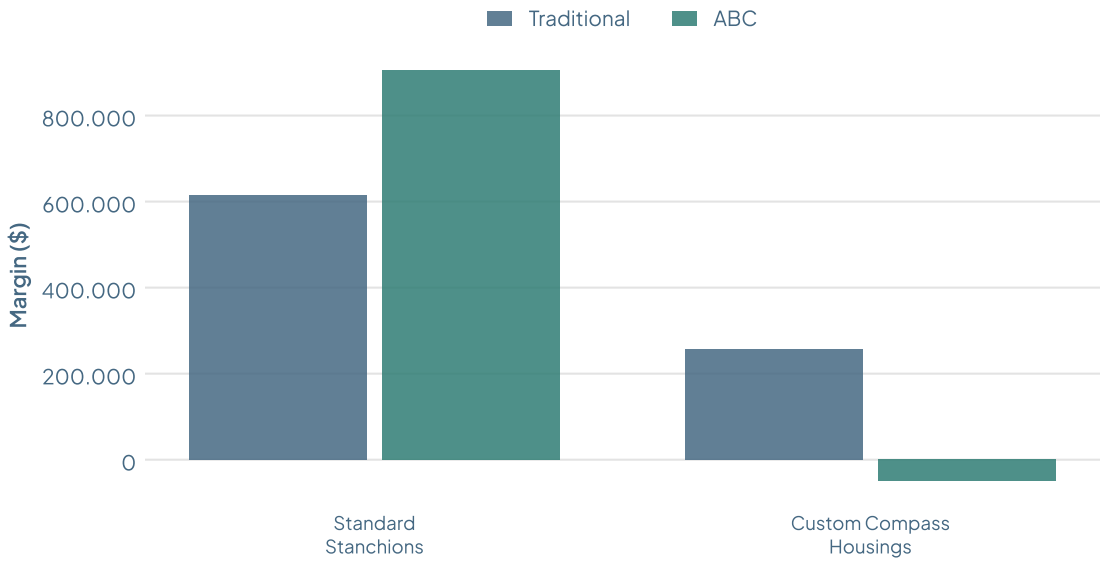
## 7 ABC in one picture: where overhead really comes from

Deep dive

Resources → Activities → Activity drivers → Products / customers

Traditional costing uses one driver (machine hours) for all overhead; custom products that use many orders and designs are undercosted.

**Classic Brass: product margin, traditional vs ABC**




The 'profitable' custom product is actually loss-making once design and order costs are traced.

**+$290,500**
Standard Stanchions margin after ABC

**−$307,500**
Custom Compass Housings margin (now a $49,500 loss)

**$1,470**
Customer relations cost per customer

## Cost hierarchy



## 8 Model limits: CVP assumptions, customer profitability and ABC weaknesses

**Four CVP assumptions** (slide: "CVP is powerful because it is simple — and risky for the same reason"):

1. The selling price is constant at any volume.
2. Costs are linear and can be cleanly split into variable and fixed (within the relevant range).
3. In multi-product companies, the sales mix is constant.
4. In manufacturing companies, units produced = units sold (inventories do not change).

The ones that most often "break" in real life are the sales mix (customers shift to cheaper products) and price (discounts to chase volume). So CVP is used as a *decision model*, not a guarantee.

**Customer profitability with ABC.** ABC is not only for products; activities can also be traced to customers.

| | Windward Yachts | Blue Harbor Fittings |
|---|---|---|
| Sales | $11,350 | $9,800 |
| Materials + labour + shipping | $4,228 | $3,850 |
| Activity costs (orders, design, order size, customer relations) | $6,423 | 15×$320 + 3×$630 + 40×$19 + 1×$1,470 = $8,920 |
| **Customer margin** | **+$699** | **−$2,970** |

Blue Harbor loses money not because its sales are small, but because those small sales consume 15 orders and 3 custom designs. Possible actions: set a minimum order size, charge separately for custom design, reduce the service level, or raise prices — and only as a last step consider ending the relationship.

**ABC weaknesses:** (1) expensive to build and maintain; (2) often resisted because the results differ from familiar numbers; (3) managers may wrongly assume every ABC cost is relevant to every decision; (4) does not always conform to external reporting standards, so two costing systems may be

needed; (5) interview-based allocations are subjective and can be manipulated.

**Decision discipline**

ABC costs are only *potentially* relevant. Always ask: "Will this cost actually change if we choose this alternative?" Organisation-sustaining costs (factory heating, the executive office) do not disappear when one product is dropped.

## Key takeaways

1. Contribution margin covers fixed costs, then becomes profit.
2. Break-even = Fixed ÷ unit CM (or ÷ CM ratio); target units = (Target + Fixed) ÷ unit CM.
3. Margin of safety = sales cushion before loss.
4. DOL = CM ÷ operating income; high fixed costs amplify swings.
5. A shift to low-CM products raises break-even.
6. Absorption vs variable costing differ only in fixed overhead.
7. ABC: resources → activities → cost objects; often reveals loss-making custom products/customers.

## Case studies

Case & worked answers

### CVP exercises: Java Roasters

Unit CM $18; CM ratio 40%; break-even 9,000 boxes = $405,000; target 14,000 boxes. At 12,000 boxes: MOS 3,000 = $135,000 = 25%; NOI $54,000; DOL 4.0; +15% sales → +60% NOI = $86,400 (verified).

Case & worked answers

## Classic Brass: ABC reverses product profitability

Activity rates: $320/order, $630/design, $19/MH, $1,470/customer. ABC overhead: Standard $524,500; Custom $427,500. Margins: Standard $615,750 → **$906,250**; Custom $258,000 → **($49,500)**. Traditional MH-based costing over-costed the high-volume product and under-costed the custom one, which consumes all design work. Windward Yachts customer margin: $699.

Case & worked answers

## Exercises 3 & 4: Meridian Furniture and Blue Harbor Fittings

**Meridian:** rates $150/order, $300/setup, $15/MH; ABC Chairs $540k, Desks $330k; traditional $29/MH → Chairs $696k, Desks $174k. Desks are undercosted — 75% of setups, 20% of machine hours.

**Blue Harbor:** activity cost $8,920; total $12,770; margin ($2,970). Many small orders plus 3 custom designs. Actions: minimum order size, separate design fees, lower service, reassess the relationship.

## Worked examples

Problem 1 · Easy **Break-even dan target laba**

Harga Rp50, biaya variabel Rp30 per unit, biaya tetap Rp200.000. Hitung (a) BEP unit dan rupiah, (b) unit untuk laba Rp60.000.

1. Unit CM = 50 – 30 = 20; CM ratio = 20 ÷ 50 = 40%.

2. ```
   BEP unit = 200.000 ÷ 20 = 10.000 unit; BEP Rp = 200.000 ÷ 0,40 = Rp500.000
   ```

3. ```
   Unit target = (60.000 + 200.000) ÷ 20 = 13.000 unit
   ```

**Answer:** BEP 10.000 unit (Rp500.000); 13.000 unit untuk laba Rp60.000.

Problem 2 Medium **MOS dan DOL**

Lanjutan: penjualan aktual 12.000 unit. Hitung laba, margin of safety, DOL, dan laba baru jika penjualan naik 10%.

1. CM total = 12.000 × 20 = 240.000 → laba = 240.000 – 200.000 = 40.000.
2. MOS = 12.000 – 10.000 = 2.000 unit; MOS% = 2.000 ÷ 12.000 = 16,7%.
3. 
```
DOL = CM ÷ Laba = 240.000 ÷ 40.000 = 6
```
4. Penjualan +10% → laba +6 × 10% = +60% → 40.000 × 1,6 = 64.000. Cek: 13.200 × 20 – 200.000 = 64.000 ✓

**Answer:** Laba Rp40.000; MOS 16,7%; DOL 6; laba naik ke Rp64.000.

Problem 3 Challenging **Absorption vs variable costing**

Produksi 10.000 unit, terjual 8.000 unit, overhead tetap Rp100.000. Metode mana yang labanya lebih tinggi, dan selisihnya berapa?

1. Overhead tetap per unit (absorption) = 100.000 ÷ 10.000 = 10.
2. Persediaan naik 2.000 unit. Di absorption, 2.000 × 10 = 20.000 overhead tetap "parkir" di persediaan (belum jadi beban).
3. Di variable costing, seluruh 100.000 langsung jadi beban periode ini.

**Answer:** Absorption costing lebih tinggi Rp20.000 (= kenaikan persediaan × overhead tetap per unit).

## Practice questions

Try answering first. The answer key is below each question.

**1. Price $45, variable cost $27, fixed cost $162,000. Break-even units?**

A. 3,600
B. 6,000
C. 9,000
D. 14,000

**Answer: C.** CM = 18; 162,000 ÷ 18 = 9,000.

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**2. Sales 12,000, profit $54,000, contribution $216,000. If sales rise 10%, profit rises about…**

A. 10%
B. 25%
C. 40%
D. 60%

**Answer: C.** DOL = 4, so 4 × 10% = 40%.

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**3. Production exceeds sales. Which profit is higher?**

A. Variable costing
B. Absorption costing
C. Equal
D. Cannot tell

**Answer: B.** Absorption parks some fixed overhead in inventory.

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**4. In the ABC hierarchy, machine setup per batch is…**

A. Unit-level
B. Batch-level
C. Product-level
D. Organisation-sustaining

**Answer: B.** Setup occurs once per batch regardless of units.

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**5. Why do custom products often look profitable under traditional costing?**

A. They always have higher prices
B. Overhead is spread by machine hours while custom products consume order and design activities
C. Lower material costs
D. High volumes

**Answer: B.** A single volume driver misses batch and design activities.

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**Sources**

- P3MD Week 5 - CVP & ABC_AZ.pdf
- UPDATED -Week 5– MK02.pdf